

BEYOND MEAN FOILS: AUDITING WORST-FOIL SPECIFICITY IN FROZEN CLIP REGION EXPLANATIONS

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ABSTRACT

A region can overlap a target object yet contribute more to another class. We test regions selected by Cluster-based Concept Importance (CCI) in frozen CLIP. Across COCO and VOC with two checkpoints, 41.08–64.78% of regions that pass overlap and mean-contrast checks fail against the strongest competing class. Removing competitors annotated in the image leaves 39.69–63.64% failing. We then test all eight candidate regions per image. An alternative passes the test for 6.25–7.84% of failures on COCO and 27.40–31.15% on VOC. Requiring it to preserve the original target-score drop within $\epsilon = .02$ reduces these rates to 0.16–0.98%. Available regions and target-drop tolerance constrain repair; relaxing the tolerance increases repair opportunities.

Index Terms— CLIP, explanation auditing, class specificity, constrained region selection

1 INTRODUCTION

Cluster-based Concept Importance (CCI) [2] explains CLIP [1] by grouping patches into regions and measuring how masking them changes image–text scores. We examine CCI-top1, the region whose removal most reduces the target score.

A region may contribute more to another class than to the target, even when its target drop exceeds the average competing drop. We test the strongest competitor, or worst foil, and ask whether another region can pass this test while preserving the target drop.

Wang and Wang’s max contrast [3] uses the highest original-image non-target logit; our worst foil has the largest drop after masking a region. Related work covers corpus foils [4], class-sensitive saliency [5], CLIP attribution [6, 7, 8], class contrast [9], and caption foils [10]. We control for competing objects and foil count, then examine all regions to locate constraints on repair.

2 AUDIT AND REPAIR OPPORTUNITY

Figure 1 shows candidate construction (a), one-prompt selection (b), and three-prompt evaluation of the fixed region (c).

For image x , let t be its target class and c any class. $\mathcal{R}(x)$ contains $K = 8$ candidate patch regions R . $s_m(x, c)$ is checkpoint m ’s pre-softmax image–text score, including its learned logit scale. Masking attention to R gives $\mathcal{I}(x, R)$. The drop $d_R(c)$ measures the region’s contribution to class c : original score minus intervened score. Positive values mean that masking lowers the score:

$$d_R(c) = s_m(x, c) - s_m(\mathcal{I}(x, R), c). \quad (1)$$

Superscript sel denotes “a photo of a {category}”. CCI-top1 chooses R_{CCI} , the region with the largest target drop; $\arg \max$ returns the maximizing region:

$$R_{\text{CCI}} = \arg \max_{R \in \mathcal{R}(x)} d_R^{\text{sel}}(t). \quad (2)$$

Evaluating class specificity. We evaluate the selected region with “a picture of the {category}”, “an image containing a {category}”, and “the {category}”. Index the $H = 3$ prompts by h . Their unit-norm text embeddings are $e_h(c)$ and their drops are $d_{R,h}(c)$. The mean embedding has length $n_c = \|H^{-1} \sum_h e_h(c)\|_2$, where $\|\cdot\|_2$ is Euclidean length. Dividing the mean drop by this length gives the drop for a normalized mean embedding: $\bar{d}_c(R) = H^{-1} \sum_h d_{R,h}(c)/n_c$.

A foil is a non-target class, and $\mathcal{F}_t$ is the set of all such classes. The margin $P(R, t)$ subtracts the largest foil drop from the target drop, with f indexing foils. A negative value means a foil loses more score than the target. With R implicit in the drops,

$$P(R, t) = \bar{d}_t - \max_{f \in \mathcal{F}_t} \bar{d}_f. \quad (3)$$

We call $P < 0$ a failure and a transition to $P \geq 0$ a sign repair. Target-response tables use the raw mean drop $H^{-1} \sum_h d_{R,h}(t)$.

Baseline set A contains images passing two checks: bbox precision (the fraction of patch centers inside target boxes) is at least .5, and the target drop exceeds the mean foil drop: $A = \{\text{bbox} \geq .5, \bar{d}_t > \max_{f \in \mathcal{F}_t} \bar{d}_f\}$.

For image i with target t_i , write $P_i(R) = P(R, t_i)$. Omitting R denotes CCI-top1. Failures form $F = \{i : P_i < 0\}$.

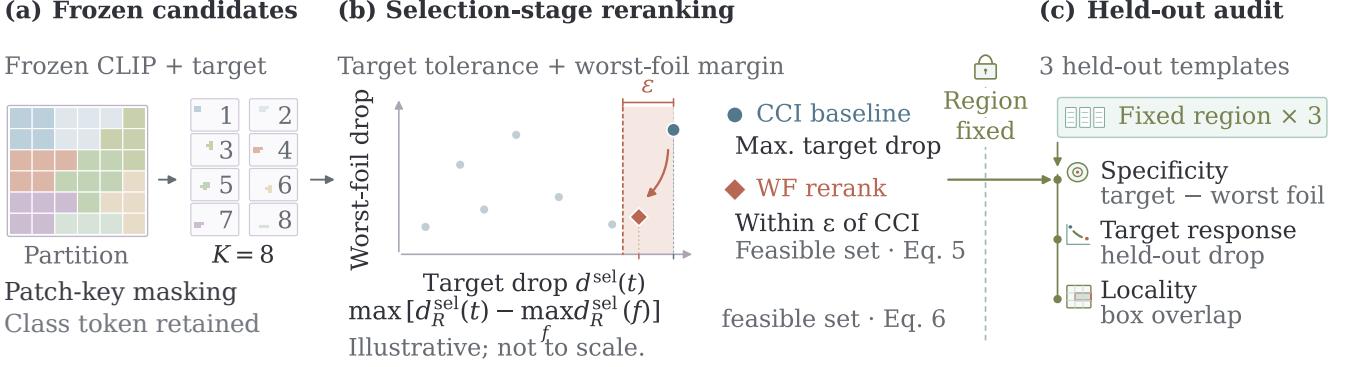


Figure 1: Selection and held-out audit. A region is fixed using selection-stage responses before three-template evaluation of specificity, target response, and locality. Schematic responses illustrate the selection rule.

The failures within the baseline set form $B = A \cap F$. Both sets stay fixed. Requiring positive target contribution gives $A^+ = A \cap \{\bar{d}_t > 0\}$.

Selecting an alternative region. Worst-foil (WF) reranking allows a target-drop loss of at most $\epsilon \geq 0$ relative to CCI-top1 (Figure 1(b)). Its selection margin M^{sel} subtracts the largest foil drop from the target drop under the selection prompt:

$$M^{\text{sel}}(R, t) = d_R^{\text{sel}}(t) - \max_{f \in \mathcal{F}_t} d_R^{\text{sel}}(f), \quad (4)$$

The feasible set $\mathcal{R}_\epsilon$ retains candidates whose target drop is at least CCI-top1’s drop minus the tolerance:

$$\mathcal{R}_\epsilon = \{R : d_R^{\text{sel}}(t) \geq d_{R_{\text{CCI}}}^{\text{sel}}(t) - \epsilon\}, \quad (5)$$

WF chooses R_{WF} to maximize that margin within the feasible set:

$$R_{\text{WF}} = \arg \max_{R \in \mathcal{R}_\epsilon} M^{\text{sel}}(R, t). \quad (6)$$

We use $\epsilon = .02$ logit units and sweep $.01/.02/.05/.10/.20$.

Locating the constraints on repair. An oracle counts images in B having at least one feasible region with a nonnegative held-out margin. Denote this count by $N_O(B, \epsilon)$ and image i ’s feasible set by $\mathcal{R}_\epsilon(i)$. The indicator $\mathbf{1}\{\cdot\}$ is one when its condition holds and zero otherwise:

$$N_O(B, \epsilon) = \sum_{i \in B} \mathbf{1}\left\{ \max_{R \in \mathcal{R}_\epsilon(i)} P_i(R) \geq 0 \right\}. \quad (7)$$

The unrestricted oracle removes the budget constraint and searches all K candidates. Each image in B falls into one of four groups: no candidate has $P \geq 0$ (C_0); a candidate passes but none meets the budget (C_1); a feasible candidate passes but WF misses it (C_2); or WF repairs the failure (r). These groups partition B , whose size is $|B|$, so $|B| = C_0 + C_1 + C_2 + r$. The oracle counts both feasible groups: $N_O = C_2 + r$. The joint condition J additionally requires overlap and mean-foil checks and $\bar{d}_t > 0$; j counts WF repairs satisfying it within fixed B . WF selects using its selection prompt; only the oracle uses held-out scores.

Controlling for competing objects. A foil may name another object in the image. To measure its contribution, remove all annotated-present classes from image i ’s foil set, leaving $\mathcal{F}_i^{\text{abs}}$. The protocol holds R_{CCI} and full-foil screen A fixed; F_i^{abs} tests $\bar{d}_t - \max_{f \in \mathcal{F}_i^{\text{abs}}} \bar{d}_f < 0$. Removing any foils can reduce failure, so we compare this exclusion with randomly retaining the same number of foils. For N_i full foils, $k_i = |\mathcal{F}_i^{\text{abs}}|$, and m_i foils with $\bar{d}_f > \bar{d}_t$, let q_i be the failure probability of a uniform size- k_i subset. With $\binom{n}{k}$ counting size- k subsets, one minus the fraction containing no stronger foil gives

$$q_i = 1 - \frac{\binom{N_i - m_i}{k_i}}{\binom{N_i}{k_i}}. \quad (8)$$

The numerator is zero when $k_i > N_i - m_i$. To isolate the effect of annotation-based removal, Δ_A averages its failure indicator minus the random-subset probability over the $|A|$ baseline images: $\Delta_A = |A|^{-1} \sum_{i \in A} (\mathbf{1}\{F_i^{\text{abs}}\} - q_i)$. Negative Δ_A favors annotation-based over random removal. Retained foil sets are nonempty.

3 PROTOCOL AND REPRODUCIBILITY

We evaluate 27,708 COCO val2014 [11] and 1,943 VOC2007 test [12] images with CLIP B/16 and B/32 (16- and 32-pixel patches): 59,302 image–model records. The target is the largest instance by annotation area (COCO) or box area (VOC). Images require another category and 2–80% target area; COCO crowd annotations are excluded. Full foils are the other 79/19 classes.

K-means uses normalized projected patch features, $K = 8$, three starts, 50 iterations, and seed $1701 + i$; i indexes sorted IDs shuffled with seed 1701. Intervention $\mathcal{I}$ adds $-\infty$ to attention logits at selected patch-key columns before softmax in every block/head, retaining the class-token column. Ties use first argmax. Boxes follow the 224-pixel short-side resize and center crop [13].

4 RESULTS AND ANALYSIS

Table 1: Fixed baseline A : full-foil, annotation-absent, and exact cardinality-matched random failure rates (%). Δ_A is absent minus random in percentage points, with paired 95% CI.

Setting	Full	Absent	Random	Δ_A
COCO B/16	64.37	63.17	64.09	-0.92 [-1.08,-0.77]
COCO B/32	64.78	63.64	64.50	-0.86 [-1.01,-0.71]
VOC B/16	42.27	40.94	41.46	-0.52 [-1.16,+0.06]
VOC B/32	41.08	39.69	40.24	-0.55 [-1.21,+0.04]

Removing annotated competitors leaves most failures. Table 1 compares the same baseline images before and after removing classes annotated in the image. The set A contains 17,726, 17,848, 1,209, and 1,227 images in table order. Failure falls by only 1.14–1.39 percentage points: 96.63–98.24% of the failures in B remain. On COCO, exclusion helps more than removing the same number of classes at random. On VOC, the confidence intervals for that difference include zero. Annotated competing objects thus explain only a small part of the observed failure.

The pattern also holds when the target contribution is positive. In A^+ , failure after exclusion remains 61.38–62.62% on COCO and 39.26–40.05% on VOC. Even COCO images with exactly two annotated categories have failure rates of 63.39% and 64.95% for B/16 and B/32, respectively. The mismatch therefore persists beyond crowded scenes.

Table 2: Exact local decomposition of *screened failures* $B = A \cap F$ at $\epsilon = .02$. Counts in C_0, C_1, C_2, r are mutually exclusive; $C_2 + r$ is the feasible-oracle count.

Setting	$ B $	C_0	C_1	C_2	r
COCO B/16	11410	10697	695	0	18
COCO B/32	11562	10655	876	1	30
VOC B/16	511	371	135	0	5
VOC B/32	504	347	154	1	2

Few failures have an alternative within the budget. Table 2 shows where repair stops. For 92.16–93.75% of COCO failures and 68.85–72.60% of VOC failures, none of the eight candidates passes the worst-foil test (C_0). Figure 2 shows the remaining opportunity as blue circles: 6.25–7.84% on COCO and 27.40–31.15% on VOC. Imposing $\epsilon = .02$ removes 96.43–98.09% of these opportunities, leaving only 0.16–0.98% of B repairable (orange diamonds). WF misses at most one of these feasible repairs per setting (C_2). The main losses occur before the final ranking decision.

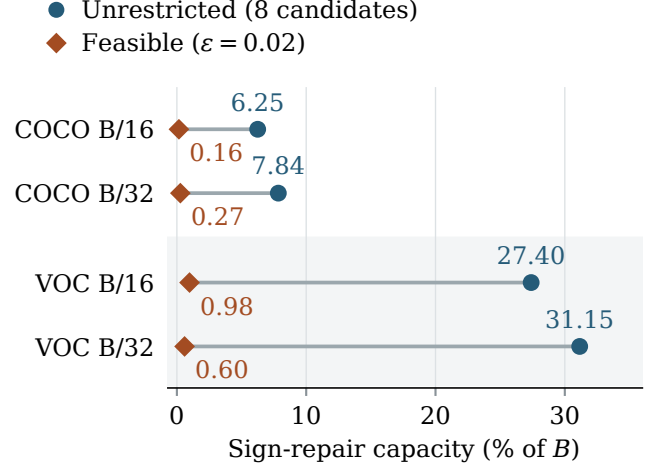


Figure 2: Budget-constrained sign-repair capacity among fixed screened failures B . Connected markers compare the same setting on a shared linear scale. Unrestricted: at least one of all eight candidates has $P \geq 0$; feasible: at least one also meets $\epsilon = .02$. Percentages use each setting’s own $|B|$ (Table 2) and are computed by exact enumeration.

Table 3: Budget endpoints (C: COCO; V: VOC). Multi: percentage of all eligible images with multiple feasible candidates. O_B, r_B : oracle/actual sign-repair percentages of screened failures B . $\Delta d_t, \Delta b$: all-image raw-target and bbox changes (bbox in percentage points).

Set	ϵ	Multi	O_B	r_B	Δd_t	Δb
C16	0.02	4.71	0.16	0.16	+0.0001	-0.12
C16	0.20	35.78	1.57	1.50	-0.0140	-1.63
C32	0.02	4.40	0.27	0.26	-0.0001	-0.08
C32	0.20	34.95	2.05	2.00	-0.0143	-0.60
V16	0.02	4.79	0.98	0.98	-0.0008	-0.34
V16	0.20	34.48	6.65	6.07	-0.0223	+0.08
V32	0.02	4.73	0.60	0.40	-0.0007	+0.02
V32	0.20	34.64	7.14	6.75	-0.0161	+1.07

A larger tolerance opens up more alternatives. At $\epsilon = .02$, about 95% of images have only one feasible candidate (Table 3). Increasing the tolerance to .20 gives multiple choices on about 35% of images. Oracle repair capacity rises to 1.57–2.05% of COCO failures and 6.65–7.14% of VOC failures. The oracle-repair percentage O_B and achieved WF percentage r_B are computed as $O_B = 100N_O/|B|$ and $r_B = 100r/|B|$.

The extra choice changes what the selected region captures. The table’s Δd_t and Δb are WF minus CCI changes averaged over all eligible images; bbox changes are in percentage points. At .20, raw target drops decrease in all four settings, while overlap decreases on COCO and increases on VOC. More repair opportunities therefore do not imply a uniform improvement in target response or localization.

Table 4: Normalization sensitivity on fixed earlier-run A_{old} and earlier-run CCI regions. Raw/Norm: failure rates (%); Flip: percentage whose failure status differs. A_{old} denotes earlier-run screen-passers.

Setting	$ A_{\text{old}} $	Raw	Norm	Flip
COCO B/16	17,726	64.318	64.369	0.107
COCO B/32	17,847	64.762	64.778	0.073
VOC B/16	1,209	42.184	42.266	0.083
VOC B/32	1,227	41.157	41.157	0.000

Normalization rarely changes the outcome. Table 4 compares raw and normalized scores while holding the earlier-run regions and images fixed. Failure rates differ by at most 0.083 percentage points, and at most 0.107% of images switch between passing and failing. The worst foil is unchanged on 99.42–99.84% of images. For example, COCO B/16 moves from 64.318% to 64.369% failure, while VOC B/32 stays at 41.157%. The high failure rate is therefore present under both score definitions.

Table 5: Actual sign repairs *within fixed baseline B*, $\epsilon = .02$. r counts repairs; j counts those also satisfying J . WF–CCI means with repair-subset 95% CIs. Δd_t is raw held-out target-score change; only bbox change is in percentage points.

Setting	r	j	Δd_t	100 Δbbox
COCO B/16	18	12	0.029 [-0.096,0.148]	-19.53 [-38.12,-1.32]
COCO B/32	30	25	0.094 [0.036,0.148]	-14.04 [-27.70,-1.74]
VOC B/16	5	3	-0.021 [-0.170,0.182]	-16.87 [-26.11,-7.00]
VOC B/32	2	1	-0.078 [-0.115,-0.042]	-28.17 [-78.57,22.22]

Repairing specificity can reduce spatial overlap. Table 5 examines only the failures that WF repairs, whereas Table 3 averages over all eligible images. Among these repairs, mean bbox precision falls by 14.04–28.17 percentage points. Mean target-score changes are positive on COCO and negative on VOC. Passing the worst-foil test therefore does not guarantee that the new region retains the original region’s overlap or target response.

The counts r and j make this distinction explicit. On COCO B/32, WF repairs 30 failures, but only 25 also satisfy the overlap, mean-foil, and positive-target checks in J . The same inequality $j < r$ holds in every setting. Even the feasible oracle finds only 12, 26, 3, and 2 repairs satisfying all checks in table order. Changes can also go the other way: among all baseline passes on COCO B/32, four become worst-foil failures. These regressions are outside the fixed failure set B .

Table 6: Nonzero-threshold sensitivity on fixed baseline A . Entries are percentages of screen-passers A with $P < -\delta$, where $\delta \geq 0$ is the required failure magnitude in the normalized margin. Thresholds are evaluated post hoc in logit units.

Setting	$\delta = 0$	.01	.10	.50
COCO B/16	64.37	63.94	59.49	36.05
COCO B/32	64.78	64.38	60.22	35.19
VOC B/16	42.27	41.44	36.23	13.73
VOC B/32	41.08	40.26	34.39	11.90

The failures are not confined to margins near zero. Table 6 tightens the test from $P < 0$ to $P < -\delta$. At $\delta = .10$, 59.49–60.22% of COCO screen-passers and 34.39–36.23% of VOC screen-passers still fail. Rates decline further at .50. Thus many failures survive a stricter margin requirement, although their prevalence depends on the chosen threshold.

Implications for region selection. The comparisons point to two distinct changes. Table 2 identifies images that need a better candidate region (C_0) and images that already have a passing region outside the budget (C_1). Figure 2 shows how sharply the budget narrows the available choices, and Table 3 shows the effect of relaxing it. Changing the ranking rule alone can recover only the additional feasible opportunities in C_2 . Improving repair therefore requires attention to which regions are proposed and how much target-drop loss is allowed.

Reproducibility and uncertainty. The primary tables come from one run that saved candidate masks, selection and evaluation responses, and bbox records together. Validation checks reconstruct the regions, scores, choices, and failure counts. The released records include candidate summaries, checks, configuration, and hashes; masks and full responses remain local. Table 4 uses earlier-run records. Three images changed selected regions between runs.

We obtain confidence intervals from 10,000 image-level bootstrap samples, using the 2.5th and 97.5th percentiles without multiplicity correction. Paired effects are differenced within each image. Foil comparisons use fixed A or A^+ ; repair-cost intervals use only actual repairs. We use exact q_i values, checked against 100 seeded random draws.

5 CONCLUSION

A region can overlap the target and beat the average competing class while contributing more to a single competitor. This occurs frequently for CCI-top1, even after annotated competitors are removed. Checking every candidate shows why reranking repairs so few failures: many images have no passing alternative, and most remaining alternatives exceed the target-drop tolerance. Relaxing that tolerance makes more repairs possible but also changes target response and overlap. Worst-foil comparisons therefore add information that overlap and mean contrast alone do not provide when evaluating frozen CLIP region explanations.

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COMPLIANCE WITH ETHICAL STANDARDS

This work analyses publicly released COCO and PASCAL VOC images and annotations. No participants were recruited and no new human or animal experiments were conducted.